



# Constructing Cybersecurity Knowledge Graph from Unstructured Data

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Metode Rekayasa Perangkat Lunak KOMB

## Kelompok 7

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# Latar belakang

## A. Permasalahan

Informasi cybersecurity saat ini tersebar di berbagai sumber seperti:

- News & security blogs, Vendor security advisories, Threat intelligence reports, Vulnerability databases (CVE)

### Permasalahan

### Dampak

Informasi tersebar di banyak sumber

Subjektif, tidak konsisten, sulit direproduksi

Data berbentuk teks bebas

Tidak dapat diproses langsung oleh sistem

Relasi antar entitas tidak terlihat

Sulit melakukan investigasi ancaman

## Tujuan Penelitian

1. Membangun CSKG dari data cybersecurity tidak terstruktur
2. Menambahkan sumber data cybersecurity yang beragam
3. Mengekstrak entitas dan relasi penting
4. Merepresentasikan hasil dalam RDF berbasis STIX
5. Menyimpan graph pada Virtuoso dan menyediakan query SPARQL

## B. Solusi

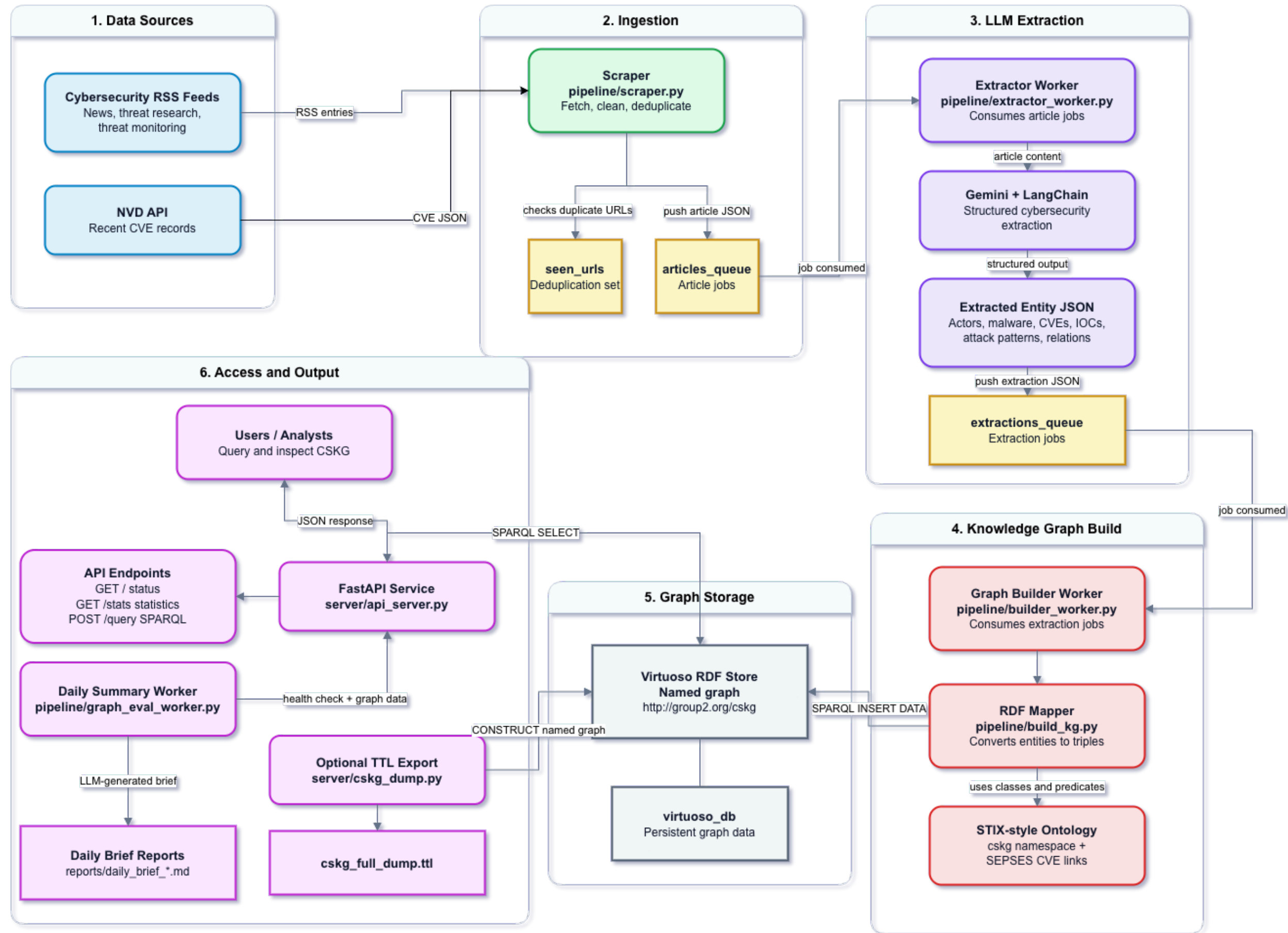
Cybersecurity Knowledge Graph (CSKG) mengubah informasi tidak terstruktur menjadi graph yang menghubungkan:

- Threat Actor, Malware, Vulnerability (CVE), Attack Pattern, Indicator, Report

Sehingga data menjadi:

- Terstruktur, Mudah ditelusuri, Dapat di-query dengan SPARQL

# Arsitektur / metode



# Implementasi

Proyek ini melibatkan pembangunan pipeline untuk mengambil data cybersecurity dari berbagai sumber, mengekstrak entitas dan relasinya menggunakan LLM, dan mengubah hasil ekstraksi menjadi RDF triples yang disimpan dalam Cybersecurity Knowledge Graph.

## 1. Data Sources

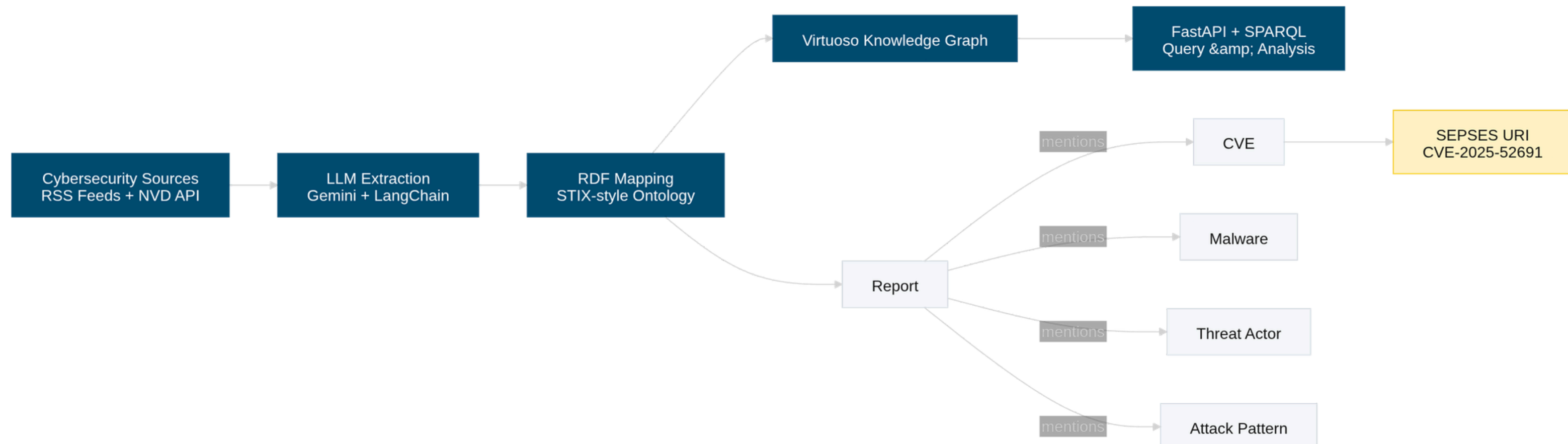
TheHackerNews, BleepingComputer, KrebsOnSecurity, Securelist, Check Point Research, SANS ISC, FortiGuard Labs, dan NVD API.

## 2. Ontology

STIX-style ontology dengan class utama: Report, ThreatActor, Malware, Vulnerability, Indicator, dan AttackPattern.

## 3. SEPSES Linking

CVE hasil ekstraksi dipetakan ke URI SEPSES. Contoh: CVE-2025-52691 → [w3id.org/sepses/resource/cve/CVE-2025-52691](https://w3id.org/sepses/resource/cve/CVE-2025-52691)



# Demo / Hasil Pengujian

Runtime evidence menunjukkan API online, graph tersimpan, dan query SPARQL berjalan.

|                              |                            |                              |                              |
|------------------------------|----------------------------|------------------------------|------------------------------|
| <b>1009</b><br>CSKG triples  | <b>94</b><br>reports       | <b>30</b><br>vulnerabilities | <b>25</b><br>malware         |
| <b>85</b><br>attack patterns | <b>60</b><br>threat actors | <b>11</b><br>indicators      | <b>11</b><br>SEPSES CVE URIs |

## Use-case SPARQL

- KG Statistics: 7 rows
- SEPSES CVE Linking: 13 rows
- Attack Pattern / Malware Context: 241 rows

## Interpretasi

Graph berhasil dibangun, dapat di-query, dan merepresentasikan hubungan antar entitas cybersecurity secara sistematis.

## Runtime Evidence

```
open.org/cti/ns/stix#Indicator":11},"count_by_predicate":
{"http://www.w3.org/1999/02/22-rdf-syntax-ns#type":305,"http://docs.oasis-
open.org/cti/ns/stix#mentions":218,"http://www.w3.org/2000/01/rdf-
schema#label":188,"http://purl.org/dc/terms/created":94,"http://docs.oasis-
open.org/cti/ns/stix#targets":49,"http://docs.oasis-
open.org/cti/ns/stix#uses":41,"http://www.w3.org/2002/07/owl#sameAs":30,"http://d
ocs.oasis-open.org/cti/ns/stix#exploits":20,"http://docs.oasis-
open.org/cti/ns/stix#reports":11,"http://docs.oasis-
open.org/cti/ns/stix#patched":9,"http://docs.oasis-
open.org/cti/ns/stix#impersonates":7,"http://docs.oasis-
open.org/cti/ns/stix#located_in":7,"http://docs.oasis-
open.org/cti/ns/stix#disrupted":6,"http://docs.oasis-
open.org/cti/ns/stix#propagated_via":5,"http://docs.oasis-
open.org/cti/ns/stix#aligned_with":4,"http://docs.oasis-
```

GET /stats — final runtime snapshot

```
"label",
"report"
],
"results": [
{
  "cve": "https://w3id.org/sepses/resource/cve/CVE-2025-52691",
  "label": "CVE-2025-52691",
  "report": "https://fortiguard.fortinet.com/outbreak-alert/smartertools-smartermail-rce"
},
{
  "cve": "https://w3id.org/sepses/resource/cve/CVE-2026-20223",
  "label": "CVE-2026-20223",
  "report": "https://thehackernews.com/2026/05/cisco-patches-cvss-100-secure-workload.html"
},
{
  "cve": "https://w3id.org/sepses/resource/cve/CVE-2026-0257",
  "label": "CVE-2026-0257",
  "report": "https://thehackernews.com/2026/05/pan-os-globalprotect-authentication.html"
},
{
  "cve": "https://w3id.org/sepses/resource/cve/CVE-2026-0257",
```

POST /query — SPARQL result: CVE, label, report



# Kesimpulan

CSKG berhasil mengubah sumber siber tidak terstruktur menjadi graph yang terukur dan dapat dianalisis.

## Berhasil

Pipeline end-to-end dari RSS/API → extraction → RDF triples → Virtuoso → FastAPI/SPARQL.

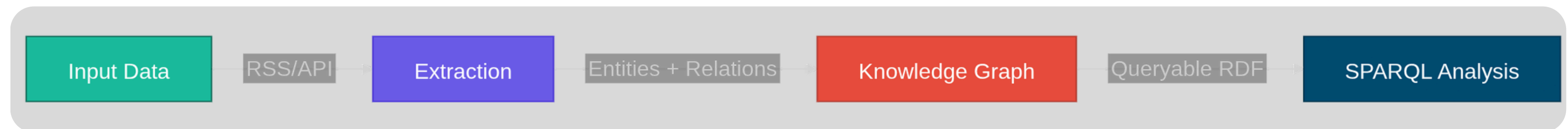
## Terukur

Final snapshot: 1009 triples, 94 reports, 30 vulnerabilities, 25 malware, 85 attack patterns, 60 threat actors, dan 11 SEPSES CVE URIs.

## Future Work

Perluasan data source, entity normalization/disambiguation, alignment ke MITRE ATT&CK/CWE, dan visualisasi graph interaktif.

## Ringkasan Alur Nilai



CSKG menjadi fondasi threat intelligence berbasis graph yang dapat dicari, dihubungkan, dan dianalisis secara sistematis.